MAT 115E Introduction to Programming Language

Lab-7/ CRN : 23492

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1 Question 1

Suppose that an array consists of positive numbers. Write a C program that reads the such an array and determines how many different numbers exist in this array.

Example Scenario:
Input: [8,5,9,5,3,5,9]
Output: 4

Test Data:
Array1=$[4, 4, 4, 4, 4, 4, 4, 4, 4]$ Array2=$[5, 12, 5, 6, 3, 4, 9, 8, 1, 8, 9]$

2 Question 2

Write a C program that accomplish the following:

- Generate an array with 12 integer elements.
- Each element of the array must be randomly generated between -20 and 20.
- Find the **SUM** of negative and positive numbers.
- Find the **NUMBER** of odd and even numbers, separately.
- Reshape this generated array into a 3×4 matrix by filling it row by row (left to right).

Example Scenario:
Input:
$[-4, 7, 5, -12, -14, 8, 9, 1, 0, 3, -10, -7]$

Output:

- Sum of negative number:-47, Sum of positive number:43
- Number of odd numbers:6, Number of even numbers:6
- $\begin{bmatrix} -4 & 7 & 5 & -12 \\ -14 & 8 & 9 & 1 \\ 0 & 3 & -10 & -7 \end{bmatrix}$

3 Question 3

Definition: In linear algebra, a symmetric matrix is a square matrix that is equal to its transpose, i.e. $A^T = A$ where A^T is the transpose of A.
Write a C program that reads square matrix of any size and check whether this matrix is symmetric or not.Your program must follow the steps below:

- You should find and print transpose of matrix you read in program.
- Then you should check these two matrix(matrix itself and its transpose) are equal to each other.